

Integrated Method for Power System Dynamic Simulation and Visualization in WECC

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Abstract—Dynamic and inconsistent loads are the hardest to plan for and the hardest for our power system to handle. The modern era introduces several new challenging loads into the grid, specifically, electrified highways and data centers. These new loads can have very large swings over very short periods of time, swings that we cannot match on the generation side, posing a hazard to the frequency of the power system. To address these problems, we must simulate and visualize the issue. This paper reviews a Julia implementation that leverages existing software tools to create a visualization of frequency dynamics while one of these problematic loads is connected to the WECC-179 bus system.

Index Terms—Grid Stability, Electrified Highways, Data Centers, ANDES, SUMO, Total Harmonic Content, WECC-179

I. INTRODUCTION

THE US power system is one of, if not the most, complicated machine in the modern era. With grid interconnections ranging from British Colombia to Florida, over one hundred and sixty million customers are attached to this sprawling network. Historically, loads have been very predictable, with linear ramp rates, consistent coincident loads, and decades of research, all working together to create an extremely reliable system (99.9% uptime according to the DoE). Rapid development in the Renewable Energy, Electric Vehicle (EV), AI, and Data Center (DC) fields have introduced new, and inherently harder to predict loads to the grid, forcing us to reconsider how we view grid dynamics.

There has not been much study into how these new loads can effect the system as a whole, and simulation of these complex systems can be intimidating. In this paper, we explore how to integrate several modern simulation tools to effectively visualize these new challenges to grid stability.

II. LOAD SIMULATION METHODS

A. Electrified Highways

As the world transitions to Electrified Vehicles the need for more sustainable infrastructure will be needed. Electrified highways (EHs) are roadways equipped with embedded transmitter coils that enable dynamic wireless power transfer (DWPT) to electric vehicles as they travel along it. As vehicles move along the roadway, power is transferred intermittently based on the overlap between the vehicle receiver and the roadway coils. Because of the periodic spacing of these coils and the continuous movement of vehicles, the resulting load demand shows time varying behavior that can create fluctuations in power consumption. This behavior is quantified as Total Harmonic Content (THC), a measure of signal distortion

that can lead to grid instability, equipment overheating, and reduced power quality.

Understanding how different traffic conditions influence these power load profiles is important for evaluating the potential impact of EH systems on power system stability. Changes in traffic flow can affect both the overall power demand and how much it varies over time, potentially introducing unwanted load behavior on the power system.

For our analysis, load profiles were generated under different traffic conditions to study the impact of key variables like vehicle speed, traffic density, and speed variability. These variables were selected to represent realistic changes in traffic behavior and to isolate their individual effects on the overall power system performance.

Vehicle trajectories used to compute load demand were generated using the Simulation of Urban MObility (SUMO) traffic simulator. SUMO provides an accurate traffic modeling framework that captures individual vehicle behavior, enabling realistic simulation of traffic flow under user controlled conditions. Detailed modeling assumptions and simulation parameters are described in later sections.

B. Data Centers

AI is infamously energy intensive, with some new data center developments having planned peak loads of 1.5 GW; over half the summer peak load of the city of Indianapolis. Not only do these new large loads put strain on our generation and transmission infrastructure, but they perform unlike any other load we have seen before, especially during LLM training cycles.

LLM training cycles typically utilize 100% of a DC's compute capacity during the weight generation phase, and approximately 30% during the weight averaging phase. These cycles typically range from 0.1-5 Hz, for this simulation we will be using 1 Hz. The duty cycle of these oscillations is always >50 , for this example we will be using a 70% Duty Cycle.

Using this piecewise unit-step approach makes a lot of assumptions about the DC's operation, but it serves as an effective worst-case scenario for dynamics simulation.

These sharp transitions from weight generation to weight averaging periods are unique, and hazardous to the grid. Transitions of this magnitude and at this speed have only one real precedent, when a generator faults and goes offline. In recent history only a handful of these faults have occurred, recently the 2021 Texas Winter Storm Blackout, and the 2019 UK Frequency Event come to mind. In both these scenarios,

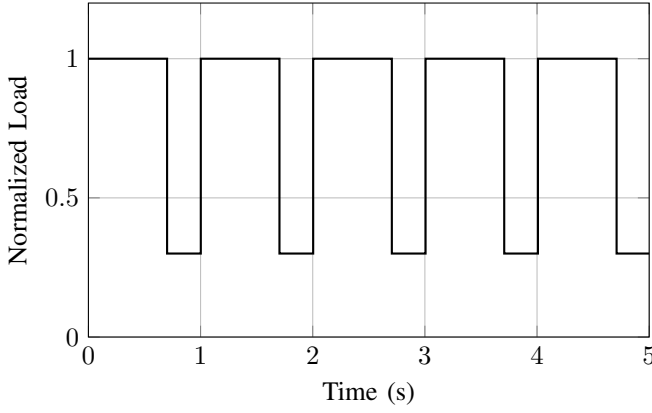


Fig. 1. Simulated DC Load

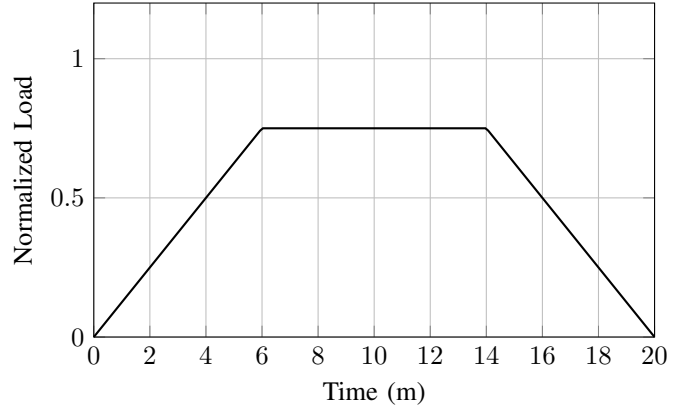


Fig. 2. Piecewise linear industrial load profile with startup ramp, steady-state operation, and shutdown ramp.

weather events caused massive, sudden loss of generation, triggering frequency collapse, and massive cascading blackouts effecting millions of people.

Data centers introduce the hazard for a similar frequency collapse, but on the load side of the system. If these swings are large enough, they could cause a large enough frequency disturbances to trip over/under frequency protection devices, and cause large blackout events.

To compound the issue, DCs demand extremely high quality power, and are capable of switching from grid generation to local generation in seconds. If this happens, the entire load of the data center can be shed immediately. This happened in Northern Virginia in 2024 causing a -1478 MW load change in a single 5 minute interval.

All things considered, DCs introduce large challenges to our grid, and we must plan for the worst to increase overall grid reliability, and avoid any potential massive blackouts.

C. Ramping Loads

Ramping loads represent the typical large load in the power system. Industrial loads like factories typically draw a lot of current with their use of rotating machinery. These loads cannot simply switch on, but they need time to ramp up, they stay relatively constant during operation, and then slowly ramp down.

As the accuracy of simulation is not the focus of this paper we will be using a simple piecewise linear function to model a ramping load. This is adequate to illustrate the difference between the irregular DC load and a regular industrial load.

III. ELECTRIFIED HIGHWAY MODELING AND ANALYSIS

A. Methodology

Multiple small traffic simulations were developed using the SUMO simulator to evaluate the impact of different traffic conditions on dynamic wireless power transfer (DWPT) load behavior. To examine the effects of specific traffic variables, a single vehicle type representing a midsize passenger car was used throughout the study. While this assumption is not fully representative of real world mixed traffic, it allows for clearer observation of how changes in speed, density, driver behavior and speed variability impact power demand.

The DWPT system was modeled based on the Indiana Department of Transportation (INDOT) testbed model. Transmitter coils of length 3.66 m were spaced by a 0.91 m gap, resulting in a total segment length of $D=4.57$ m. Each vehicle was assumed to be equipped with a battery receiver with length 2.0 m. The instantaneous power delivered to each vehicle was computed based on the overlap between the vehicle receiver and the roadway transmitter coils. Overlap from adjacent coil segments were summed and scaled by a power density factor of 109.36 kW/m. The power density factor represents the amount of power delivered per unit overlap length and was chosen to approximate realistic DWPT system performance, with adjustments to reflect specific modeling assumptions.

The first roadway was modeled as a straight 4 km highway segment consisting of two lanes, where only the right lane was electrified. Vehicle trajectories were generated using small timesteps of 0.01 seconds over a total simulation time of 600 seconds. To eliminate transient effects with initial traffic population, only data from 200 to 600 seconds were used for analysis.

At each timestep, the total system power demand was calculated by summing the instantaneous power of all vehicles in the electrified lane. Traffic conditions were varied across three different parameters: density, speed, and speed variability. Traffic density was controlled by varying the vehicle arrival rate using a Poisson distribution, where parameter λ represents the average number of vehicles entering the roadway per second (veh/s). Values of λ were 0.25, 0.5, and 0.75 veh/s, which were used to represent low, medium, and high traffic density conditions, respectively.

Vehicle speed was controlled using a scaling parameter called a speed factor in SUMO. A value of 1.0 corresponds to the standard speed of 74 mph, while values of 0.8 and 1.2 were used to represent lower and higher speed conditions. In practice the achieved speeds varied slightly due to traffic interactions. Speed variability was introduced by allowing individual vehicle speeds to vary around the average speed. Variability levels of 5%, 10%, and 20% were used to represent low, medium, and high variability conditions. But in practice vehicle traffic interactions limited the variability measured in

the simulation, as faster vehicles were constrained by slower leading vehicles.

The second roadway was modeled as a straight 2 km highway with 2 lanes, where again only the right lane was electrified. This simulation collected data from 200 to 1000 seconds with the last car entering the roadway at 1000 seconds. In this simulation at about the 260 second mark a car crashes in the EV lane to simulate what could be a real world scenario. This crash happens at 1.4 km from the left side of the highway. With this scenario Vehicles per hour, and tau (driver behavior) were all changed between multiple value to see how it effects the THC.

The third roadway developed was developed to see how a 4 way light could effect a similar highway. In a real world scenario the DWPT would not be implemented in big cities roads with many stop light, but they could be implemented on a highway that features one or two stoplights over many miles. This simulation collected data from 0 to about 1000 seconds, when the last car exits, with the last car entering the roadway at 800 seconds. This 4 way light featured 2 way traffic from both North-South and East-West. The East-West segments of road has 1 km on each side of the light and the North side has 200 meters and south has 500 meters. The junction has a 70 second cycle with the East-West having a priority green light with 42 seconds with 3 seconds of yellow. In this scenario the cars can randomly turn left or turn right if they are in the respective lanes. In this scenario the rightmost lane of both sides of the East-West 2 way traffic has a electrified highway lane. This scenario was only ran once with one set of parameters.

To measure how power varies over time, the average power $\bar{p}$ was computed from the total instantaneous power $p(t)$, where $p(t)$ represents the total power demand of all vehicles in the electrified lane at time t .

$$\bar{p} = \frac{1}{T} \int_0^T p(t) dt \quad (1)$$

The fluctuation magnitude E was computed as the root-mean-square deviation of $p(t)$ from the average:

$$E = \sqrt{\frac{1}{T} \int_0^T (p(t) - \bar{p})^2 dt} \quad (2)$$

A normalized fluctuation metric, defined as the total harmonic content (THC), was used to compare variability across different scenarios:

$$\text{THC} = \frac{E}{\bar{p}} \quad (3)$$

This metric compares fluctuations in power to the average level. To observe how THC varies over time, it was evaluated using a 30 second moving window throughout the primary simulation period. No additional smoothing or filtering was applied, so the results reflect the natural behavior of the system.

B. Results

1) *Traffic Density on highway 1:* The impact of traffic density on system performance was evaluated by varying the

vehicle arrival rate. The resulting simulation produced densities of 5.65, 9.82, and 13.86 veh/km, which we considered as low, baseline, and high density.

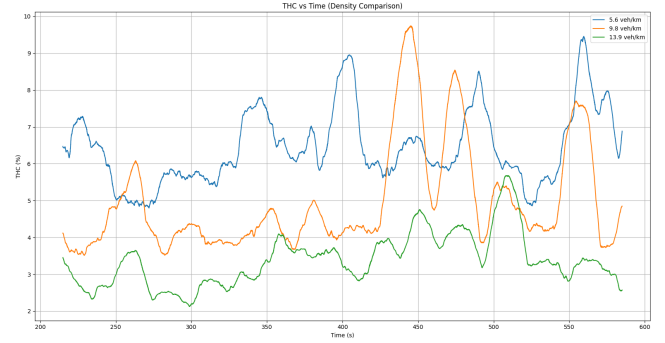


Fig. 3. THC versus time for three traffic density cases: 5.65, 9.82, and 13.86 veh/km. While the fluctuation patterns are roughly similar across all cases, lower density results in higher THC magnitudes while the higher density results in lower overall values.

The time varying THC results are shown in Fig. 3. A clear trend is observed, where lower traffic density results in higher THC values, while higher density leads to reduced fluctuations. The least dense case has the largest and more prominent peaks, while the most dense case remains mostly stable over time, with smaller fluctuations than the other test cases. Although occasional peaks overlap between the different cases, the overall separation between density levels is consistent throughout the simulation.

The density test cases power profiles are shown in Fig. 4. In contrast to the THC behavior, higher density results in higher overall power levels, with the most dense case maintaining the highest power throughout the simulation. The power remains relatively steady over time for each case, with only moderate fluctuations compared to the changes observed in THC.

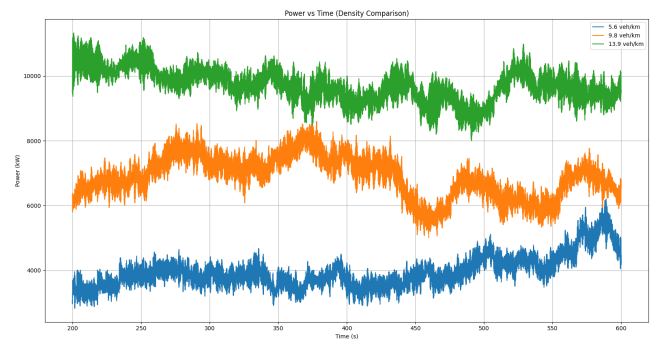


Fig. 4. Total power load (kW) versus time for three traffic density cases: 5.65, 9.82, and 13.86 veh/km. Higher traffic density results in increased overall power levels, while all cases have similar fluctuation patterns and maintain relatively stable power profiles over time.

The overall THC values for the low, baseline, and high density cases were 12.8%, 9.39%, and 5.19%, respectively. This shows a strong inverse relationship between traffic density and load fluctuations. Where increasing the number of vehicles leads to smaller relative power fluctuations, as shown by the decrease in THC across the different test cases.

2) *Speed With Constant Density on highway 1*: The impact of vehicle speed on the power system was evaluated under similar traffic density. Three cases were tested, corresponding to measured average speeds of 46.8 mph, 59.2 mph, and 70.2 mph, with densities of 10.7, 9.7, and 10.2 veh/km, respectively.

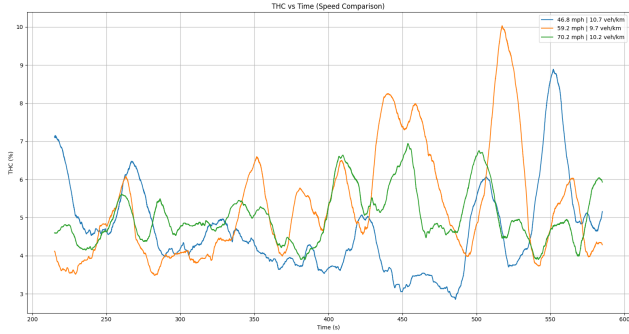


Fig. 5. THC versus time for three vehicle speed cases: 46.8 mph, 59.2 mph, and 70.2 mph. The corresponding traffic densities are approximately 10.7, 9.7, and 10.2 veh/km, respectively. Despite differences in speed, the THC profiles have similar overall behavior with variations mainly in the timing and magnitude of fluctuations.

The time-varying THC results are shown in Fig. 5. All three cases show fluctuations over the simulation period, with noticeable peaks and valleys. The lower and baseline speed test cases have more pronounced spikes, while the higher speed case appears more stable over time, with fewer extreme variations. Despite these differences in behavior, all cases follow similar general trends, indicating that speed alone does not significantly alter the overall pattern of THC over time.

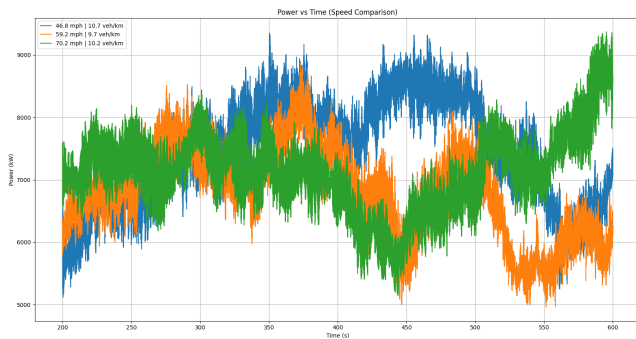


Fig. 6. Total power load (kW) versus time for three vehicle speed cases: 46.8 mph, 59.2 mph, and 70.2 mph, with comparable traffic densities. The overall power levels are similar across all cases, with differences primarily in the timing and variability of fluctuations. Slight deviations are observed across all cases toward the end of the time window.

The corresponding power profiles are shown in Fig. 6. The power levels are similar across all three cases, with differences primarily occurring in the timing and magnitude of fluctuations. Periods of rapid change in power tend to align with peaks and valleys in THC over time, while more stable power regions are associated with lower THC values.

The overall THC values for the slow, baseline, and high speed cases were 11.01%, 10.51%, and 8.98%. This means there is a slight decrease in overall fluctuation as speed

increases, although the effect remains relatively modest across the tested range.

3) *Speed Variation With Constant Density on highway 1*: The impact of speed variability on system performance was tested while maintaining constant traffic density across the separate test simulations. Three cases were considered, with intended variability levels of 5%, 10%, and 20%. The resulting measured variability levels were 4.1%, 6.7%, and 6.1%, with corresponding densities of 9.33, 9.82, and 9.46 veh/km, respectively.

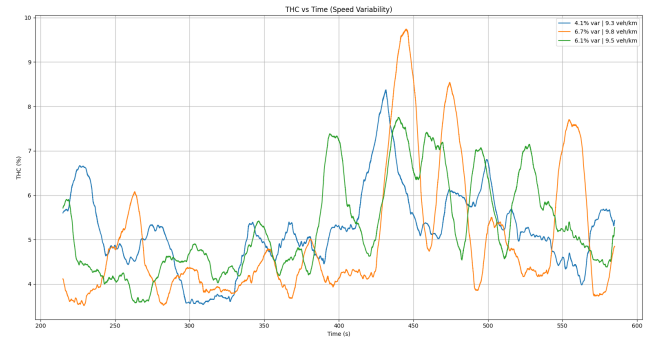


Fig. 7. THC versus time for three speed variability cases with target standard deviations of 5%, 10%, and 20%. The measured speed variations are 4.1%, 6.7%, and 6.1%, with corresponding traffic densities of 9.3, 9.8, and 9.5 veh/km. Despite these differences the THC profiles remain similar, indicating limited impact to speed variability.

The time varying THC results are shown in Fig. 7. All three cases showed similar fluctuation patterns over time, with the peaks and valleys occurring at nearly the same intervals. While the medium variability case shows slightly larger spikes at certain times, the overall behavior across all three cases remains consistent. Across all the cases there was no clear separation between variability levels.

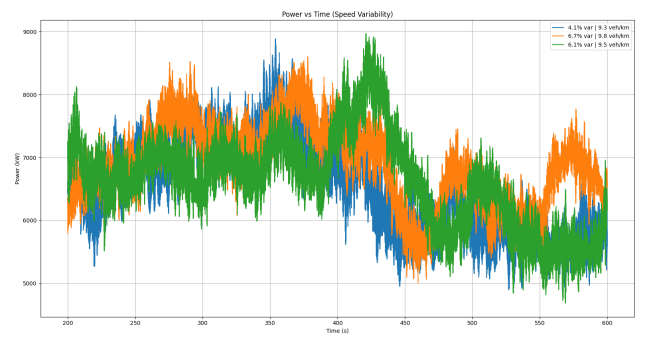


Fig. 8. Total power load (kW) versus time for three speed variability cases is measured at 4.1%, 6.7%, and 6.1%. The power profiles are similar across all cases, with comparable overall levels and only minor differences in fluctuations over time.

The separate power profiles are shown in Fig. 8. The power graph for all three cases largely overlap, indicating small differences in overall power behavior as variability changes. Any slight differences that do appear are minimal and do not persist over time.

The overall THC values for the low, baseline, and high variability cases were 11.44%, 9.39%, and 10.98%, respectively.

These results are close together and show only minor differences across the tested conditions. This suggests that speed variability has a limited impact on overall power fluctuations under the given simulation model.

4) *Vehicles per hour on highway 2:* Below you can see the plots of both THC and power consumption for the Vehicle per hour tests. These plots were achieved by having a constant Tau of 1.2 and a speedFactor=normc(1.05,0.1,0.9,1.2). This means the lowest a car's speed factor will be is 90% the speed limit, the highest is 120%, the mean is 105%, and the standard deviation is .1. With a speed limit of 25 meters per second or 56 miles per hour. In this simulation vehicles per hour was changed from 1500-3500 in 500 vehicle per hour segments.

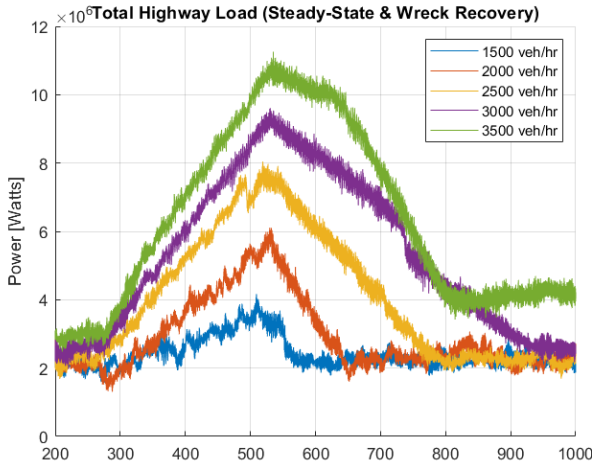


Fig. 9. Total power load (W) versus time for five vehicle per hour cases: 1500 Veh/H, 2000Veh/H, 2500Veh/H, 3000Veh/H, 3500Veh/H with comparable speed factors and tau.

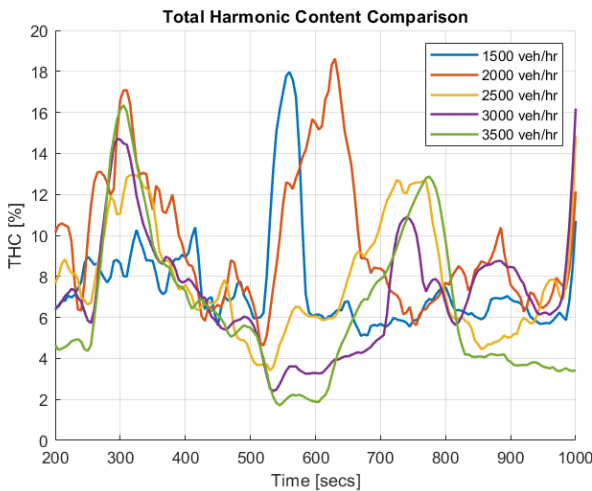


Fig. 10. THC versus time for five vehicle per hour cases: 1500 Veh/H, 2000Veh/H, 2500Veh/H, 3000Veh/H, 3500Veh/H with comparable speed factors and tau.

As seen in the power graph there is a linear relationship between number of cars and total demand. The 3500 VPH scenario reaches a peak load of approximately 11 MW, while

the 1500 VPH case peaks at only 4 MW. This is obviously due to more vehicles on the road drawing more power. At 3500 VPH, despite the massive increase in total power consumption, the THC dropped to its lowest recorded value of approximately 2%. This saturation effect occurs because the high density of vehicles fills the gaps between charging coils, smoothing the signal into a more continuous DC load. In contrast, the 2000 VPH scenario seemed to be the worst, sustaining peak THC of nearly 18% because it lacks the necessary saturation to smooth the individual power pulses generated by vehicles.

5) *Driver behavior on highway 2:* Below you can see the plots of both THC and power consumption for the tau tests. Tau is a variable that affects driver behavior, mainly by affecting the following distance of drivers. With a tau of 1 that means a driver follows 1 second behind the driver in front of them. This can also make them more aggressive to lane swaps and cutting people off. These plots were achieved by having a constant vehicle per hour of 2500 and the same speed factor as before. In this simulation tau was changed from 1 to 1.2 to 1.5 to 1.8.

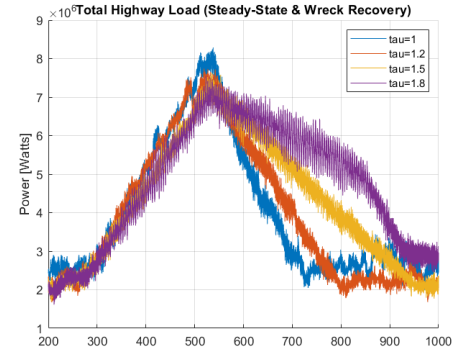


Fig. 11. Total power load (W) versus time for four tau cases: tau=1, 1.2, 1.5, 1.8 with comparable speed factors and Veh/Hour

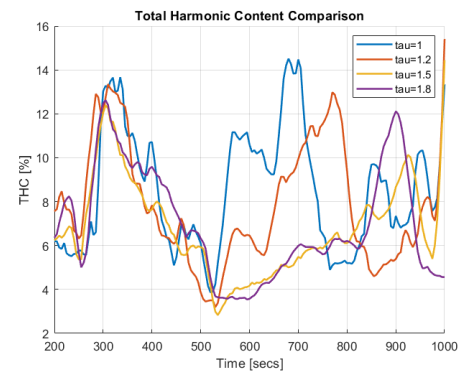


Fig. 12. THC versus time for four tau cases: tau=1, 1.2, 1.5, 1.8 with comparable speed factors and Veh/Hour

The analysis of driver behavior within the highway 2 network shows a distinct correlation between tau and the stability of the electrical grid. Specifically, the results indicate that THC decreases significantly as the tau increases from tau=1.0 to tau=1.8. You can also see that the later peak due to cars clearing out tends to be later with the higher tau. Aggressive

drivers ($\tau=1.0$) produced the most severe harmonic distortion with the THC constantly hitting peaks, with THC peaks reaching approximately 15% during the incident recovery phase. Conversely, conservative drivers ($\tau=1.8$) maintained a significantly flatter and more stable harmonic profile, typically hovering near 6%. Aggressive drivers exited the recovery zone quickly causing the power to lower quicker, while conservative drivers exhibit a more sustained power consumption stretching the load profile across a longer time interval.

6) *Highway 3 Results:* No more cars enter after 800s so data from 200-800 is most relevant. Below you can see the Figure 13 which has the total power load for this scenario and Figure 14 which has the THC for this scenario.

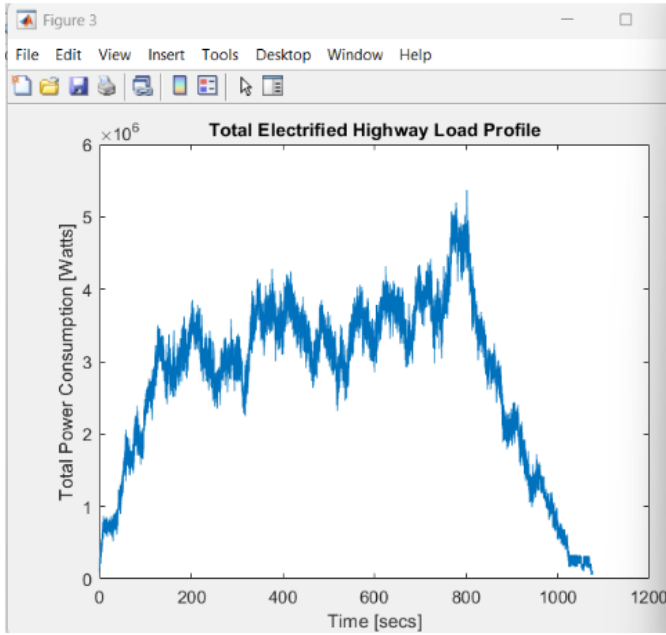


Fig. 13. Total power load (W) versus time for highway 3 (4-way light)

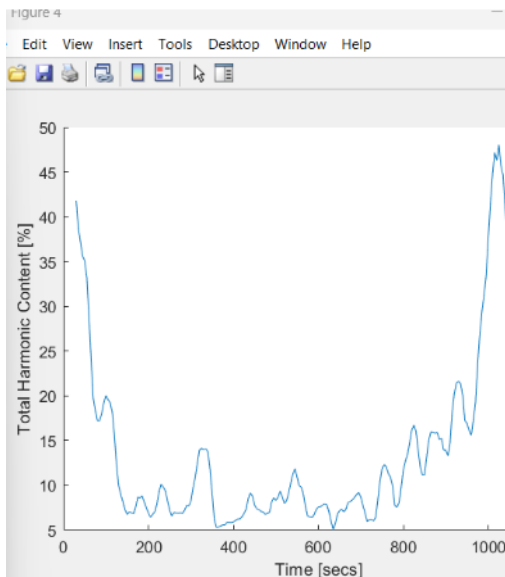


Fig. 14. Total Harmonic Content versus time for highway 3 (4-way light)

Every time the light turns green, the queued EVs accelerate simultaneously. This results in massive power spikes peaking near 5 MW as multiple vehicles hit the charging slabs at the exact same moment. You can also see peaks in the THC peaking at about 15 at these same times. When the light is red you can see that the Power and the THC drop to its lowest points. This is due to the cars being stationary and drawing a constant load.

C. Discussion

1) *Traffic Density:* The results show a clear relationship between traffic density and power variability. As density increases from 5.65 to 13.86 veh/km, THC drops from 12.8% to 5.19%. This shows that higher vehicle density leads to a more stable overall power demand, with less fluctuations over time. This trend is visible in the THC vs. time plot, where the lower density case shows larger and more frequent spikes, while the higher density case remains more stable.

At the same time, the power vs. time results show the opposite trend, where higher density creates more overall power consumption. But this is expected, as more vehicles are drawing power from the electrified roadway the overall load will be greater.

These observations can be explained using theoretical scaling relationships. Let N be the number of vehicles drawing power from the electrified lane. Under realistic traffic conditions where vehicles are not synchronized, the fluctuation component of the load scales with $\sqrt{N}$, while the total power demand scales with N . As a result, the THC scales with the number of vehicles as

Density has a strong and predictable impact on power demand, with increased traffic leading to more stable power delivery. The big takeaway is that higher traffic density improves power stability by reducing fluctuations, even though it increases total power demand.

$$\text{THC} \propto \frac{1}{\sqrt{N}} \quad (4)$$

This relationship indicates that as more vehicles draw power, their individual fluctuations tend to average out. This smoothing effect explains why higher density scenarios exhibit lower normalized variability despite having a higher total power demand.

2) *Speed With Constant Density:* The results demonstrated that vehicle speed has a limited impact on power load fluctuations when traffic density is held approximately constant. The corresponding densities for the three cases were 10.7, 9.7, and 10.2 veh/km, having only minor variation across the different tests. As speed increases from 46.8 mph to 70.2 mph, THC decreases from 11.01% to 8.98%. Although there is a slight trend, the overall change is minor compared to the stronger effect of density. This implies that speed alone does not significantly influence the stability of the power load under the tested conditions.

From a power demand viewpoint, the overall power levels remain similar across all speed cases. There are small differences in load fluctuations, but their overall magnitudes are

very similar. Times of rapid change in power still align with increases in THC, but this kind of behavior is consistent across all speed conditions.

This behavior can be explained by considering how speed affects the system. Vehicle speed determines how frequently a vehicle interacts with the roadway coils, which changes the timing of power transfer over time. At higher speeds vehicles pass over each coil more quickly, leading to more frequent but shorter power transfer events, while at lower speeds these interactions occur less often but last longer. But when the density of vehicles is similar at different speeds, the total level of power demand on the road doesn't change significantly. Because of this, the overall effect remains similar, and the fluctuations do not change much with speed.

When compared to just density, which directly changes the number of contributing vehicles, speed primarily affects the timing of power transfer rather than its overall variability. This explains why changes in speed produce only minor differences in THC, while varying density remains a prominent factor in determining power load stability.

3) *Speed Variation With Constant Density*: The impact of speed variability on power system performance was evaluated while maintaining similar traffic density across all test cases to isolate its effects. The measured densities for the three cases were 9.33, 9.82, and 9.46 veh/km, showing little variation across tests. Variability levels of 5%, 10%, and 20% were set for each model, but the measured variability values were much lower, at 4.1%, 6.%, and 6.1%, respectively, indicating that the intended range of variability was not fully executed in the simulation.

Because of the similar outputs, the resulting THC values were roughly the same, with values of 11.44%, 9.39%, and 10.98% for the low, baseline, and high variability cases. The THC vs. time graph showed similar fluctuation patterns across all cases, with no clear separation between variability levels. The power profiles from each simulation largely overlap, indicating minimal differences in overall system behavior.

This behavior can be explained by the effects of vehicle interactions within the traffic flow. Even though we set individual vehicle parameters, faster vehicles are often constrained by slower leading vehicles. This limits faster vehicles' ability to execute their intended speed variability. As a result the effective spread in vehicle speeds is reduced from the intended parameter, leading to similar system behavior across all cases.

From a power perspective variability in individual vehicle speeds mostly affects the timing of power transfer rather than the overall magnitude of fluctuations. Since the number of vehicles and their spacing remain similar, the power demand is not significantly changed. This explains why changes in speed variability produce only minor differences in THC, despite larger intended variations in individual vehicle behavior.

D. Conclusion

The results show that traffic density has the most significant impact on power load fluctuations, with higher densities producing more stable power profiles and lower THC values. Furthermore, changes in vehicle speed result in only slight differences, mainly affecting the timing of power transfer rather

than its overall variability. Speed variability was found to have a minimal impact under the tested conditions, largely due to limited effective variation in vehicle speeds. These findings indicate that density is the dominant factor in determining system stability, while different speeds and speed variability have a smaller impact.

IV. DYNAMICS SIMULATION METHOD

For power dynamics simulation we will be employing the Automatic Numerical Differential-algebraic Equation Solver (ANDES) [1] tool produced by the University of Tennessee's CURENT Large Scale Testbed. This tool allows for detailed and validated power dynamics simulations. As our focus is on the effect of isolated loads on a large network, we will be simulating using the WECC-179 bus system. This system approximates the Western Interconnection with buses from British Columbia to New Mexico.

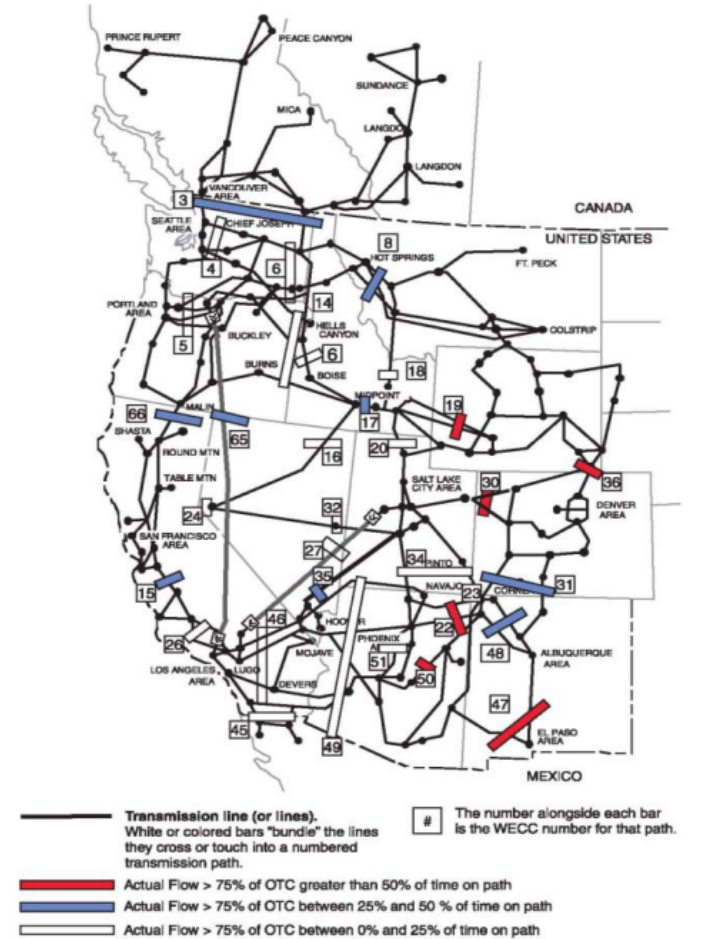


Fig. 15. WECC map

These ANDES simulations output normalized frequency data at each bus, we use this data to visualize power system dynamics. ANDES is mostly used for dynamics simulations given one constant grid state, or few states including a fault. However, by leveraging ANDES' alter function, we can insert any load shape, formatted as a time series, to any bus. For our purposes we simply select one load bus, and insert our

load shape of choice (SUMO Data, DC waveform, ramping waveform) and execute the following code:

```
begin
  for (i,bus_id) in enumerate(bus_conc)
    for (j, t_event) in enumerate(time)
      p_val = powers[j]
      ss_conc.add("Alter", Dict(
        "model" => "PQ",
        "dev" => "LLM$(i+1)",
        "src" => "Ppf",
        "attr" => "v",
        "method" => "=",
        "amount" => np.float64(p_val),
        "t" => np.float64(t_event)
      ))
    end
  end
end
```

V. VISUALIZATION OF POWER GRID FREQUENCY DYNAMICS USING GEOSPATIAL HEATMAPS

The increasing integration of large-scale data centers into power systems introduces highly dynamic and periodic load profiles, which can significantly impact grid stability. In this project, we develop a visualization framework to analyze and interpret frequency deviations across a power grid under such dynamic loading conditions. Using time-series simulation data from a Western Electricity Coordinating Council (WECC) system model, we generate a geospatial heatmap that dynamically illustrates frequency deviations across multiple nodes. The visualization provides insight into how disturbances propagate spatially and temporally across the grid. This approach enables intuitive understanding of complex grid dynamics and supports future research in grid stability under emerging load conditions such as data centers and electrified transportation.

A. Introduction

Modern power systems are increasingly exposed to non-traditional loads such as data centers, whose power consumption can vary rapidly due to computational workload distribution. These variations can introduce periodic disturbances (1 second timescale) that affect grid frequency stability.

While numerical simulations can capture these effects, interpreting raw time-series data across multiple buses is difficult. Therefore, the objective of this work is to:

Visualize frequency deviations across the grid Capture spatial propagation of disturbances Provide an intuitive representation of dynamic grid behavior

This project focuses on transforming simulation outputs into a dynamic geospatial visualization of frequency deviations.

B. Methodology

1) *Data Source and Processing*: Simulation data was obtained using the ANDES power system simulator applied to a Western Electricity Coordinating Council (WECC) network

model. The dataset consists of time-series values of generator angular frequencies across multiple buses.

The frequency deviation at each node is computed as:

$$\Delta f = (\omega - 1.0) \times 60 \quad (5)$$

where ω represents the per-unit angular frequency and 60 Hz is the nominal grid frequency. This conversion provides frequency deviation in Hertz, which is more meaningful for analysis.

The dataset was processed using Julia, where CSV files were imported into DataFrames, relevant generator frequency columns were extracted, and deviations from nominal frequency were computed. The data was also normalized to ensure consistent visualization scaling.

2) *Geospatial Mapping*: Each generator bus in the WECC system was mapped to approximate geographic coordinates defined by latitude and longitude values. These coordinates allow the simulation data to be projected onto a real-world geographic map.

A subset of buses representing key regions of the grid was selected to ensure clarity while maintaining sufficient spatial coverage. These nodes serve as anchor points for spatial interpolation.

3) *Spatial Interpolation*: Since frequency measurements are available only at discrete nodes, a spatial interpolation method is required to generate a continuous field across the region.

A Gaussian radial basis function (RBF) was used for interpolation:

$$W_i(x, y) = e^{-\gamma((x-x_i)^2 + (y-y_i)^2)} \quad (6)$$

The interpolated frequency field is computed as:

$$f(x, y) = \frac{\sum_i W_i(x, y) \Delta f_i}{\sum_i W_i(x, y)} \quad (7)$$

where (x_i, y_i) represents the coordinates of node i , Δf_i is the frequency deviation at that node, and γ is a tuning parameter that controls the smoothing effect. As γ increases, the interpolated radius decreases. A smaller γ means the reach of the interpolation is smooth and broad, while a larger one means it is trying to wrap close to individual data points. The ideal γ was calculated using a cross validation program across the data points, and averaging the highest scores. [2] This method ensures smooth spatial transitions while maintaining local influence of each node.

4) *Visualization Framework*: The visualization was implemented using Julia-based libraries including CairoMakie and GeoMakie. Geographic boundaries were incorporated using Natural Earth datasets to provide spatial context.

The framework includes the following components:

- Heatmap representing interpolated frequency deviation
- Scatter points indicating actual node values
- Geographic overlays including state boundaries and coastlines
- Time-evolving animation using observable variables

The color scale is centered around zero, allowing clear differentiation between positive and negative frequency deviations. The final output is exported as a time-dependent video illustrating the evolution of grid dynamics.

C. Code Implementation

The visualization pipeline was implemented in Julia using libraries for data processing and geospatial plotting. The most important components of the code are summarized below.

First, the simulation data is loaded from a CSV file and converted into a usable format. The generator angular frequency values are extracted and converted into frequency deviation from the nominal 60 Hz using:

$$\Delta f = (\omega - 1.0) \times 60 \quad (8)$$

This step is critical because it transforms raw simulation outputs into physically meaningful values that represent grid stability.

Next, each generator bus is mapped to a geographic coordinate (latitude and longitude). This allows the frequency data to be placed onto a real-world map. Only a subset of buses is used to maintain clarity while still representing the overall grid behavior.

Since the data is only available at discrete bus locations, interpolation is required. The code uses a Gaussian weighting approach, where nearby nodes have a stronger influence on each point in the grid. This creates a smooth, continuous heatmap of frequency deviation across the region.

A spatial grid is then defined over the WECC region, and the interpolated values are computed for each grid point. This is the core step that converts discrete simulation data into a visual field.

For visualization, the code uses GeoMakie to generate a heatmap of frequency deviation overlaid on a map of the western United States. State boundaries and coastlines are added to provide geographic context. In addition, the actual bus locations are plotted as points on top of the heatmap to show the original data sources.

Finally, the visualization is animated over time using an observable variable that updates the heatmap at each time step. The frames are recorded and exported as an MP4 video, showing how frequency disturbances evolve across the grid.

Overall, the key components of the code are:

- Converting simulation data into frequency deviation
- Mapping buses to geographic coordinates
- Interpolating discrete data into a continuous field
- Rendering a geospatial heatmap with overlays
- Animating the results over time

D. Results

These plots show the generator frequency outputs for each type of load. The y-axis is normalized frequency, $1.00 = 60$ Hz. It is important to note that the ANDES simulation does not adjust generation to match load, so the large shifts shown in the ramp/ electrified highway graphs can be ignored. The sharp peaks and valleys are what we are concerned with in

this paper. Evidently, the data center loads caused significantly larger frequency disturbances than either other load shape, the largest swings being 0.018 Hz. The data center simulated is 500 MW. Hardly any harmonics are evident in the ramp load, and only minor, albeit very irregular, harmonics are evident in the electrified highway output.

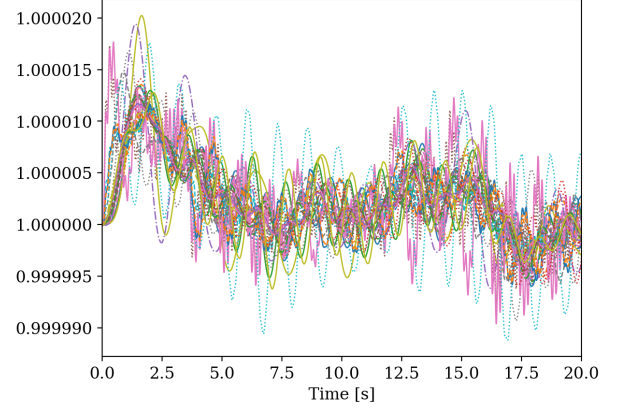


Fig. 16. Electrified Highway Frequencies

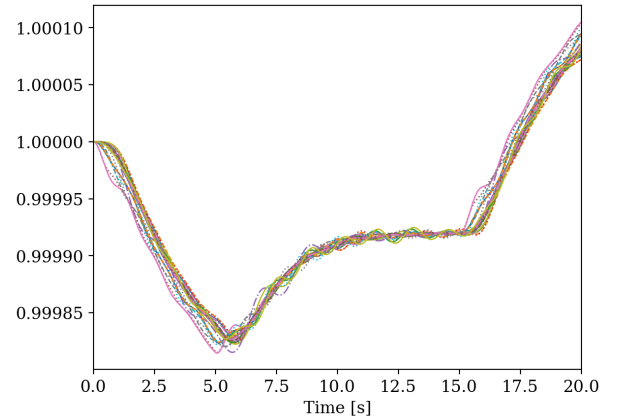


Fig. 17. Ramp Generator Frequencies

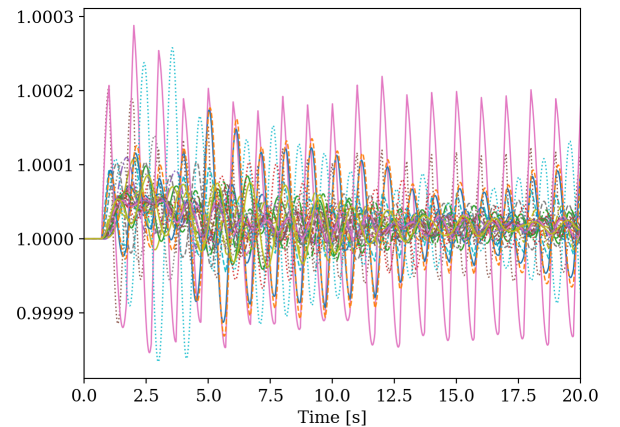


Fig. 18. Data Center Generator Frequencies

The generated visualization illustrates the temporal evolution of frequency deviations across the WECC grid under dynamic loading conditions.

The results show that frequency disturbances propagate spatially across the network rather than remaining localized. Certain regions exhibit stronger deviations due to network topology and electrical connectivity.

Oscillatory behavior is observed, consistent with periodic load variations introduced by data center operations. Early time responses appear relatively uniform, while later stages show more complex spatial interactions.

The visualization enables identification of:

- Direction of disturbance propagation
- Regions with higher sensitivity to frequency deviations
- Time delays between responses at different nodes

E. Discussion

The results demonstrate that periodic load variations, even at small time scales, can lead to significant spatial frequency dynamics in the power grid.

The grid behaves as a coupled dynamic system where disturbances propagate similarly to waves. The geographic distribution of nodes and transmission infrastructure plays a key role in determining how these disturbances evolve.

From the perspective of data center integration:

- Periodic computational loads can induce sustained oscillations
- These oscillations may interact with natural system dynamics
- Certain regions of the grid may be more vulnerable to instability

The visualization framework provides an intuitive way to analyze these effects, which would be difficult to observe using raw numerical data alone.

F. Conclusion

This work presents a visualization framework for analyzing power grid frequency dynamics using geospatial heatmaps.

The key contributions include:

- Transformation of simulation data into spatial representations
- Application of RBF-based interpolation for continuous visualization
- Development of a dynamic animation framework for temporal analysis

The results highlight the importance of understanding spatial dynamics in modern power systems, especially with the increasing presence of dynamic and non-traditional loads.

G. Future Work

Future work can extend this framework in several directions:

- Integration with real-time grid monitoring data
- Use of higher-resolution network models
- Exploration of adaptive interpolation techniques
- Application of machine learning for anomaly detection

VI. INTEGRATION

We wrote our program in Julia. Julia has a few benefits that made it best for this use case. First, it is able to run any python library, which makes it compatible with ANDES. Second, it is fully open source so the code can be shared and used by anyone free of cost. And finally, it eliminates the library dependency problem that is frequently faced with python code sharing.

The general flow of our combined Julia implementation can be seen in the below figure.

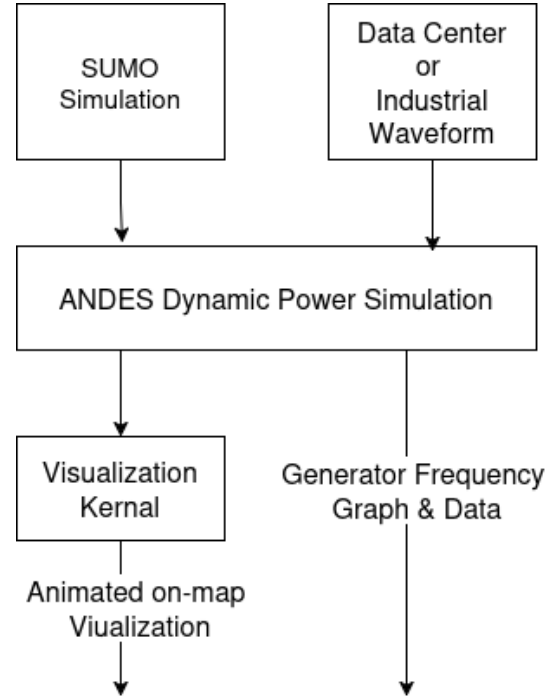


Fig. 19. Integrated Code Workflow

VII. CONCLUSION

This paper demonstrated an integrated approach to simulating and visualizing the complex dynamics introduced by modern, high-intensity loads within the WECC-179 bus system. By leveraging the Julia programming language and the SUMO traffic simulator, we successfully modeled the frequency impacts of Electrified Highways (EH) and Data Centers (DC)—two infrastructure developments that pose unique challenges to grid stability.

Our analysis highlights that the intermittent nature of Dynamic Wireless Power Transfer and the aggressive oscillations of LLM training cycles create frequency disturbances far exceeding those of traditional industrial ramping loads. Specifically, the high Total Harmonic Content (THC) associated with varying traffic densities and the sharp “step-function” power demands of data centers represent a new frontier of risk for frequency collapse.

Ultimately, these findings underscore the necessity for advanced visualization and real-time simulation tools in modern grid planning. As the power system transitions toward a more

volatile load environment, integrated methods like the one proposed here will be essential for engineers to develop robust protection schemes, optimize generation responses, and ensure the continued reliability of the North American power grid.

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